

Redefining Food Delivery with Autonomous Vehicles

A case study in research-led design for AI-powered systems

TEAM / ROLE

- Experience
Design Lead
- Research
Lead
- Product
Manager
- Engineering
Lead
- Industrial
Designer
- Dig./Phys.
Prototyper

COMPANIES

- 
Motional
- Uber
Eats

Context & Opportunity

The Situation

Motional partnered with Uber Eats for their first on-road autonomous vehicle delivery pilot. It was an opportunity to differentiate in a crowded AV space by proving we could take on both rideshare and food delivery reliably and at scale.

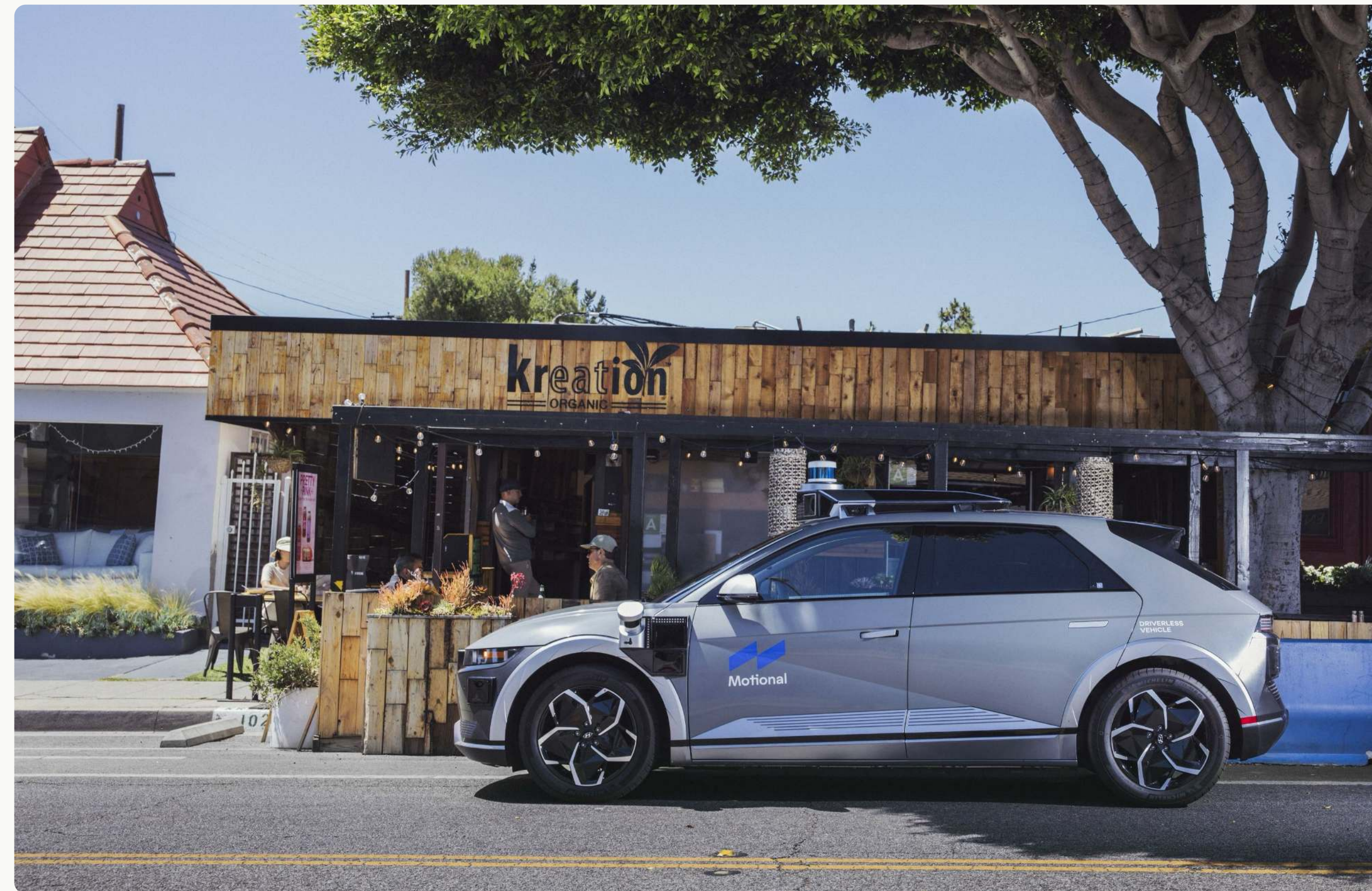
This pilot became a springboard for deeper Uber collaboration and a chance to design trust into a brand new experience.

Design Challenges:

- **AVs were unfamiliar:** How do we create trust and excitement, not confusion?
- **Tech alone doesn't solve problems:** Human behavior is messy, and we needed to design for it.

Why This Mattered to Users:

- **Merchants:** More predictable handoffs with fewer unknowns. One less person to coordinate with during busy service hours. Clear expectations.
- **Recipients:** Lower cost. Less awkwardness. Greater security and control.



Project Role & Goals



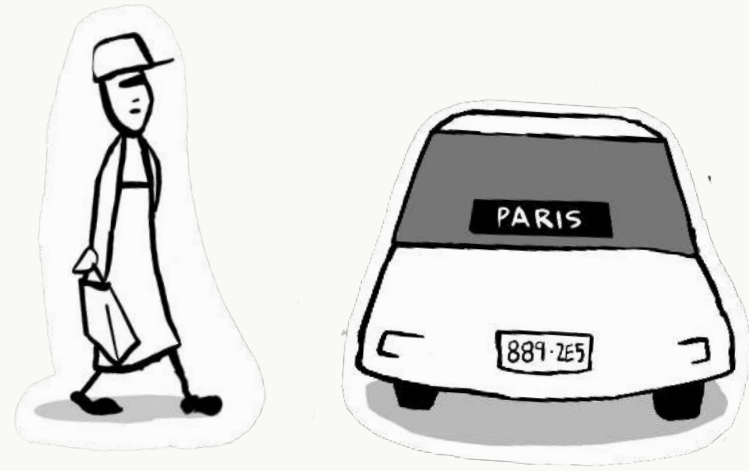
What We Set Out to Solve:

- Build a trusted, intuitive autonomous vehicle delivery experience
- Identify and reduce friction for merchants, recipients, and vehicle operators
- Inform vehicle enhancements, in-app experience, and future scale through research
- Support commercial teams with tools and insights to grow partnerships strategically

Actions Taken as Experience Design Lead:

- Advocated for on-site immersion early in the project, shifting the team from assumption-based to human-centered decision making
- Collaborated with product and research leads on a multi-perspective learning agenda
- Created storyboards and testing tools for merchant and recipient research
- Built multi-modal prototypes to evaluate usability (authentication, retrieval, handoff)
- Led design exploration of vehicle UI, compartment ID, and AV-human interaction
- Created journey maps that informed the pilot's operational design and scaling strategy

Research Strategy



Merchant & Vehicle Research

- Ridealongs + Vehicle Operator Interviews
- Authentication Testing (QR Code vs. Keypad)
- Merchant Interviews (store environments, order handoff pain points)

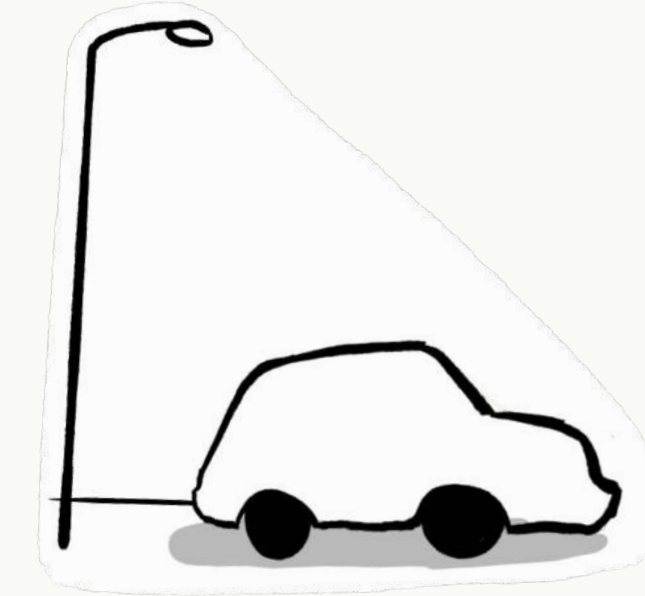
Takeaway: Merchants struggled with competing delivery platforms; authentication had usability gaps.



Eater Experience & Wayfinding

- In-home Interviews (How do people interact with an AV for food delivery?)
- Wayfinding & Order Retrieval Testing (Where do they go? Are they comfortable using it?)

Takeaway: Users needed stronger digital+physical guidance to know what to expect and avoid feeling lost or awkward in public.



Market Fit & Future Scaling

- Virtual Interviews (Power users & motivations)
- Quantitative Survey (Broader validation)

Takeaway: AVs had broader market potential, influencing vehicle storage design. Future Use Cases (Grocery? Package delivery?)

Turning Insights Into Action

Our findings and design decisions shaped how the pilot evolved, and helped broader Motional and Uber see the value of experience-led thinking in a tech-first project.

Turning Insights Into Action

The Merchant Environment

INSIGHT

Merchants juggled in-store customers, multiple delivery platforms, and noisy order screens, needing a simple, secure way to load food into the AV and return to their tasks.

DESIGN DECISIONS

We explored QR code scanning and Keypad Entry, ultimately landing on Keypad authentication using the order number on the receipt:

- No need for merchants to locate a personal device
- Eliminated issues with battery, Wi-Fi, or poor lighting
- Built trust with a low-friction, familiar format

OUTCOME

This became a hardware input requirement for future vehicles and reduced risk and merchant worries around loading errors.



Turning Insights Into Action

Setting Expectations for Recipients

INSIGHT

Recipients were auto-opted into the driverless pilot, but many didn’t realize a driverless car would arrive, or that they’d need to meet it at the curb. This led to confusion and missed connections.

DESIGN DECISIONS

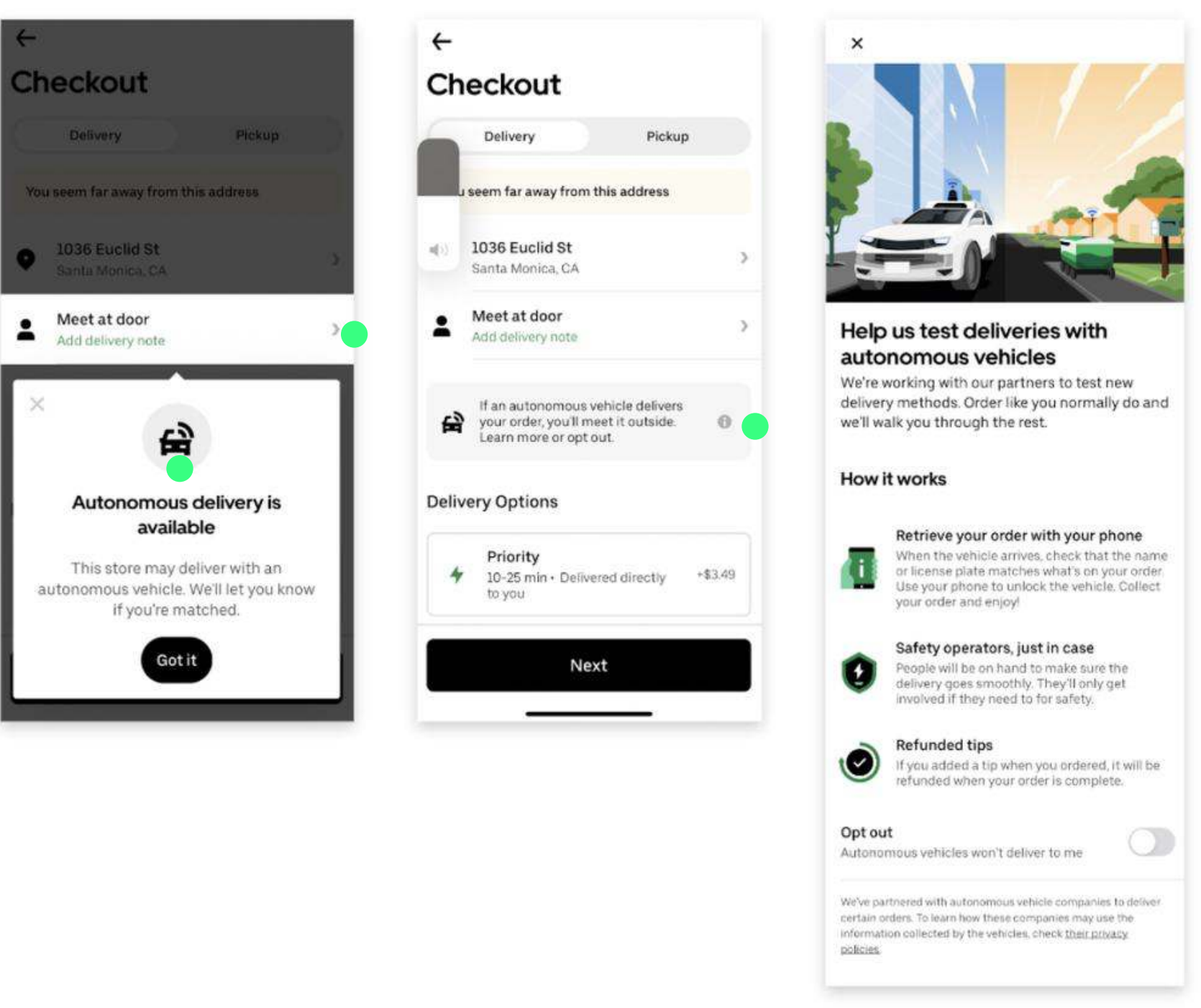
We introduced clearer pre-arrival messaging with:

- Strengthened visual cues showing an AV might arrive
- Messaging enforcing value: energy-efficient, no tip required

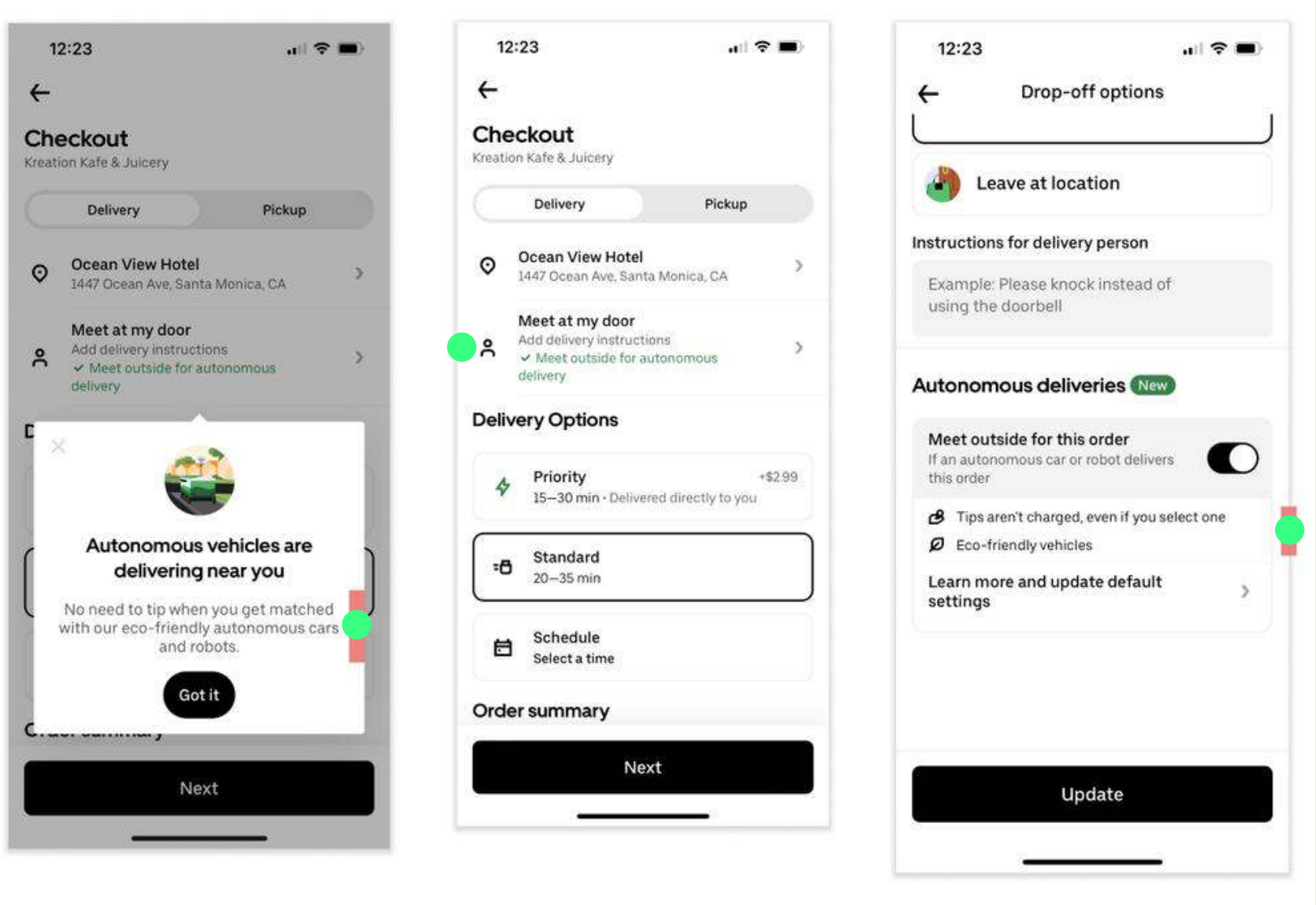
OUTCOME

Better expectation-setting helped build trust and made the AV delivery feel like a feature, not a surprise. We tracked and saw reductions in time on delivery as well as completion rate.

UE Pilot Exp V1



UE Pilot Exp V2



Turning Insights Into Action

Order Loading & Retrieval Guidance

INSIGHT

Even without a driver to ‘steal a fry’, human behavior still introduced risk: grabbing the wrong item, opening the wrong compartment, or simply being unsure how to get their order. Guidance was essential.

DESIGN DECISIONS

We identified future product requirements based on findings:

- Lockable, powered compartments with individual access
- Logic for order placement and retrieval (compartment ID)
- On-screen or in-app guidance to match order to compartment

OUTCOME

These requirements informed future model design, so that it could support self-serve retrieval securely, without needing human intervention.



Outcomes & Impact

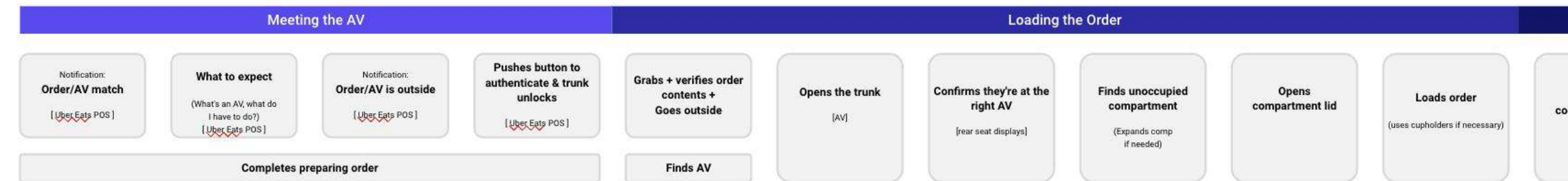
Outcomes & Impact

How Design Moved the Pilot and the Business Forward

- Reframed the project from proving tech to solving human experiences
- Created journey maps that visualized the complete merchant and recipient flow, surfacing friction points and failure scenarios that hadn't been considered in preparing for a purely driverless experience
- Enabled alignment across product, engineering, and commercial teams around what a “ready-for-scale” experience needed to look like
- Directly influenced pilot expansion to new partners like Shake Shack, helping prioritize which merchants to bring on based on real-world delivery dynamics
- Set the foundation for how Motional and Uber Eats think about scaling driverless delivery beyond the pilot



Merchant Happy Path



Assumptions

Orders will be loaded into the trunk.



Eater Happy Path



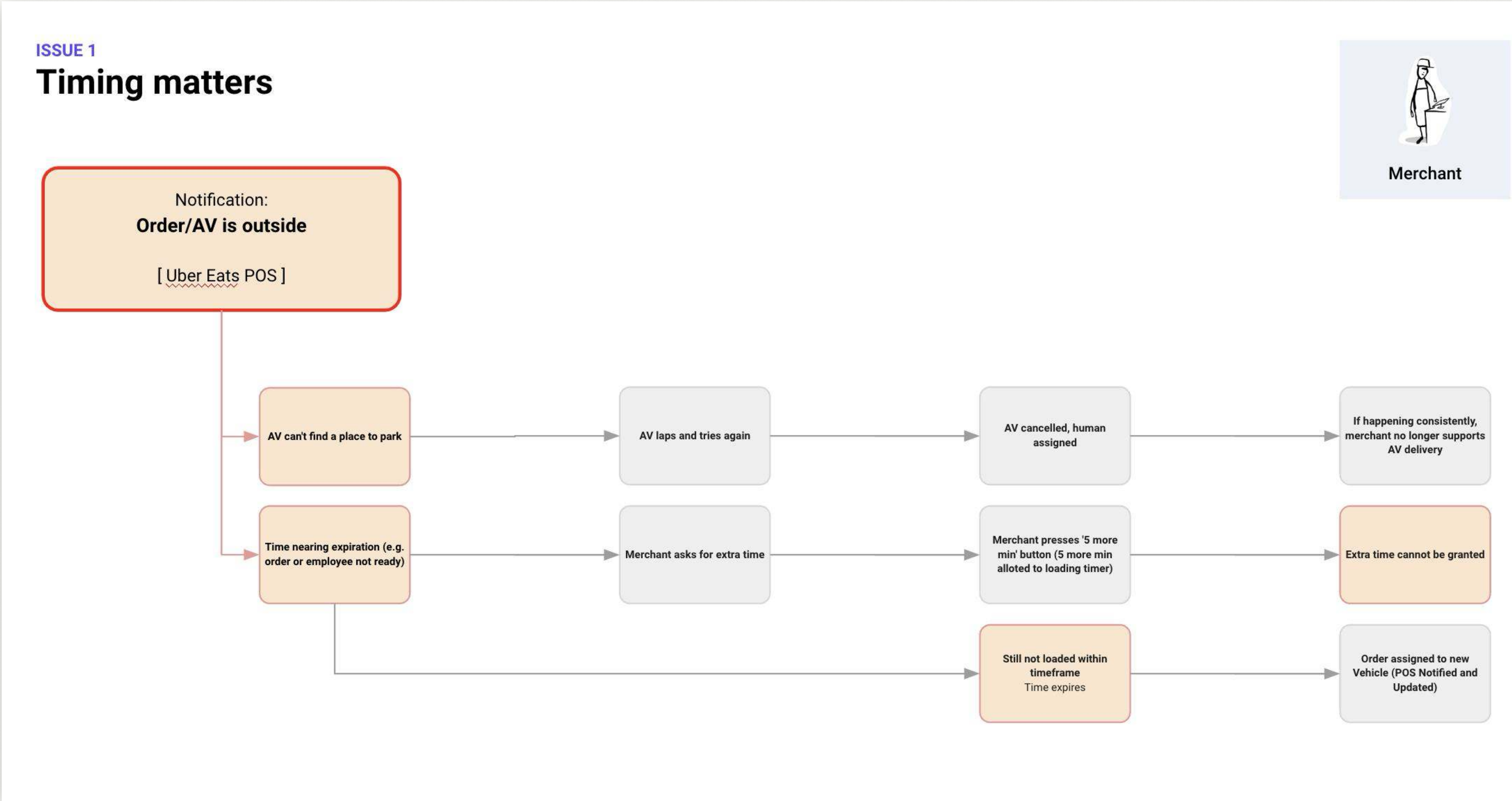
Assumptions

Orders will be unloaded from the trunk.

AV can carry 1-2 orders at a time. (Batching allowed)

2 Unpowered, non-locking compartments.

We unpacked what can go wrong after an AV arrives, revealing parking issues or merchant delays.



Proposing mitigation strategies like timers and reminders, and flagging downstream system implications.

ISSUE 1

Timing matters



Problems

- Unlike rides, neither merchants nor customers had to go outside to meet the vehicle prior to AV delivery, and may be slow/inconsistent in doing so
- Vehicle cannot sit at locations indefinitely, especially if double parked
- There are multiple situations where the user could act slowly or incorrectly, preventing the happy path flow from continuing

Unhappy Path Solutions

- Series of built-in timers that account for user not acting quickly enough, triggering some resolution for that stage in the flow
- Series of reminders and warnings sent to user interface to speed them up to avoid more serious issues

Implications of Solutions

- Multiple timers, triggers, UI warnings, and deadline outcomes will be difficult to build
- Some post-timeout resolutions are unclear
- Vehicle may or may not have order inside
 - No way to automatically check for this
 - Vehicle action should be different depending on whether or not it has food inside

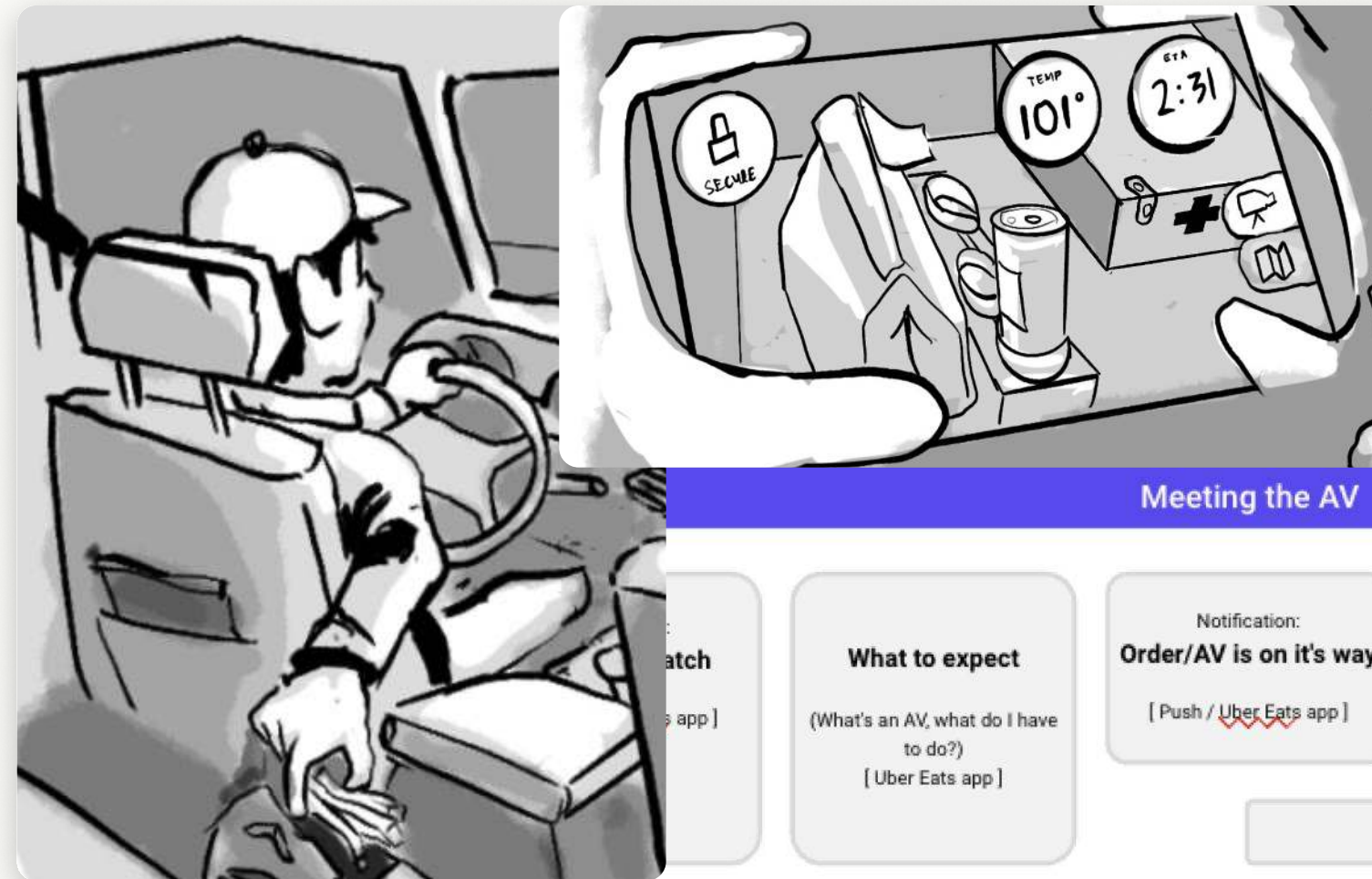


Looking Back & Looking Ahead



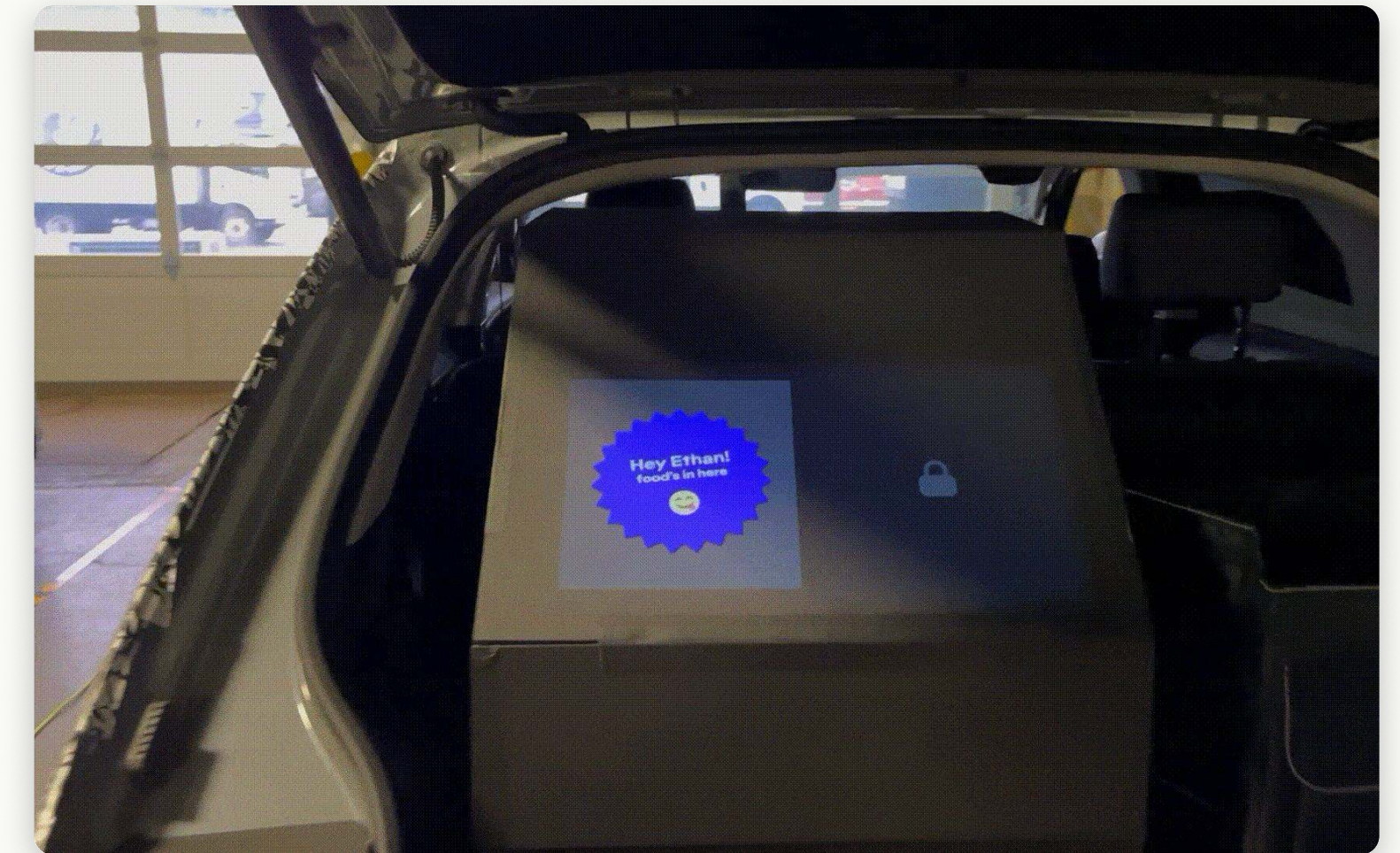
Shifting from Proving Tech to Designing for People

We made findings actionable, influencing how Uber Eats positioned autonomous delivery and refining key interactions, like improving feedback collection.



Designing for an Unknown Future

Visuals! Materialize a collective imagination. Journey mapping kept us grounded, clarifying focus and surfacing gaps/pain points.



The Power (and Limits) of Prototyping

Delays in a prototype can break the feeling of reality, especially with futuristic tech. I'd be excited to push prototyping further next time, leveraging tools that create more seamless, high-fidelity simulations.



Thank You